Identifying Key Entities in Recipe Data

Named Entity Recognition using Conditional Random Fields

Submitted by: Soumyadeep Saha

IIIT Bangalore

September 2026

Table of Contents

1. Introduction & Business Objective..... 3

2. Data Description..... 3

3. Data Ingestion and Preparation..... 4

4. Train-Validation Split..... 5

5. Exploratory Data Analysis..... 6

6. Feature Extraction for CRF Model..... 8

7. Model Building and Training..... 10

8. Prediction and Model Evaluation..... 11

9. Error Analysis..... 12

10. Conclusion..... 13

1. Introduction & Business Objective

The goal of this project is to train a Named Entity Recognition (NER) model using Conditional Random Fields (CRF) to extract key entities from recipe data. The model classifies words into predefined categories — ingredients, quantities, and units — enabling the creation of a structured database of recipes and ingredients.

This structured data can power advanced features in recipe management systems, dietary tracking apps, or e-commerce platforms for automated ingredient parsing and inventory management.

2. Data Description

The dataset is provided in JSON format, containing 285 recipe records. Each record has two fields:

Field	Description
input	Raw ingredient list from a recipe (space-separated tokens)
pos	Corresponding NER labels: ingredient, quantity, or unit

- Label Descriptions:
  - ingredient — Name of a food item (e.g., "Turmeric", "Onion", "Salt")
  - quantity — Numeric value or amount (e.g., "2", "1/2", "3")
  - unit — Measurement unit (e.g., "cups", "teaspoon", "grams")

3. Data Ingestion and Preparation

3.1 Loading and Initial Exploration

The JSON file was loaded into a pandas DataFrame using a custom load_json_dataframe() function. Initial exploration revealed:

Metric	Value
Total Records	285
Columns	2 (input, pos)
Data Types	object (string)

3.2 Tokenization and Validation

The input and pos fields were split into token lists (input_tokens and pos_tokens). Length validation identified 5 rows where input and POS token counts did not match:

Index	Input Length	POS Length
17	15	14
27	37	36
79	38	37
164	54	53
207	18	17

These 5 rows had incomplete POS annotations (each missing exactly 1 label) and were dropped, leaving 280 clean records.

3.3 Unique Labels

The unique_labels() function confirmed exactly 3 valid NER labels: ingredient, quantity, and unit. No erroneous labels were found.

4. Train-Validation Split

The cleaned dataset of 280 records was split into training and validation sets using a 70:30 ratio with random_state=42:

Split	Count
Training	196 recipes
Validation	84 recipes
Total	280 recipes

All 3 unique labels (ingredient, quantity, unit) were present in both training and validation sets, ensuring representative coverage.

5. Exploratory Data Analysis

5.1 Token Distribution

Flattened token analysis on the training set revealed the label distribution:

Label	Count	Percentage
ingredient	5323	74.8%
quantity	980	13.8%
unit	811	11.4%

Key Observation: Significant class imbalance exists, with ingredient tokens comprising ~75% of all tokens. This imbalance was addressed during model training through class weights.

5.2 Top 10 Most Frequent Ingredients (Training)

The most frequently occurring ingredients in the training data include common Indian cooking staples such as salt, onion, oil, chilli, cumin seeds, garlic, turmeric, coriander, ginger, and curry leaves. These ingredients appear across a wide variety of recipes, reflecting the dataset's focus on Indian cuisine.

5.3 Top 10 Most Frequent Units (Training)

The most common measurement units include teaspoon, tablespoon, cup, unit, sprig, cloves, grams, inch, and pinch. "Teaspoon" and "cup" dominate, aligning with typical recipe measurement patterns.

5.4 EDA Insights

- The ingredient label dominates the dataset (~75%), creating a class imbalance that needs to be addressed.
- Common Indian ingredients (salt, onion, oil, chilli) appear most frequently.
- Standard cooking units (teaspoon, tablespoon, cup) are the most common units.
- The training and validation sets show similar distributions, indicating a representative split.

6. Feature Extraction for CRF Model

6.1 Keyword Sets and Patterns

Domain-specific keyword sets were defined to aid entity recognition:

- Unit Keywords: cup, cups, tablespoon, teaspoon, gram, grams, kg, liter, ml, ounce, pinch, bunch, sprig, slice, clove, cloves, inch, etc.
- Quantity Keywords: half, quarter, one, two, three, ... ten, dozen, few, some
- Quantity Pattern (Regex): Matches fractions (1/2, 1-1/2), decimals (3.14), and whole numbers (5)

6.2 Feature Function (word2features)

A comprehensive word2features() function was designed with 25+ features per token:

Core Features:

- bias: Constant feature (1.0) to aid model learning
- token: Lowercase form of the current token
- lemma: Lemmatised form via spaCy
- pos_tag: Part-of-speech tag from spaCy
- tag: Detailed POS tag
- dep: Dependency relation in the sentence
- shape: Token shape (e.g., "Xxx" for "Milk")
- is_stop: Whether the token is a stopword
- is_digit: Whether the token consists only of digits
- has_digit / has_alpha: Presence of digits or alphabetic characters
- hyphenated / slash_present: Presence of hyphens or slashes
- is_title / is_upper / is_punct: Capitalisation and punctuation checks

Quantity & Unit Detection Features:

- is_quantity: Matches quantity pattern or keyword
- is_unit: Matches unit keyword set
- is_numeric / is_fraction / is_decimal: Specific numeric format detection
- preceding_word / following_word: Adjacent token context

Contextual Features:

- prev_token / next_token: Adjacent tokens in lowercase
- prev_is_quantity / prev_is_digit: Type of preceding token
- next_is_unit / next_is_ingredient: Type of following token
- BOS / EOS: Beginning/End of sentence markers

6.3 Class Weight Computation

To address class imbalance, inverse frequency weights were computed and the ingredient label was further penalised:

Label	Class Weight
ingredient	0.67 (penalised)
quantity	8.65
unit	8.77

7. Model Building and Training

7.1 CRF Model Configuration

Parameter	Value
Algorithm	L-BFGS
c1 (L1 regularisation)	0.5
c2 (L2 regularisation)	1.0
Max Iterations	100
All Possible Transitions	True

7.2 Training Results

Classification Report on Training Data:

Label	Precision	Recall	F1-Score	Support
ingredient	1.0000	1.0000	1.0000	5323
quantity	1.0000	0.9959	0.9980	980
unit	0.9951	1.0000	0.9975	811
Weighted Avg	0.9994	0.9994	0.9994	7114

Training Accuracy: 99.94%

The model was saved as crf_model.pkl using joblib for future use.

8. Prediction and Model Evaluation

Classification Report on Validation Data:

Label	Precision	Recall	F1-Score	Support
ingredient	1.0000	1.0000	1.0000	2107
quantity	0.9903	0.9951	0.9927	411
unit	0.9944	0.9888	0.9916	358
Weighted Avg	0.9983	0.9983	0.9983	2876

Validation Accuracy: 99.79% (2870/2876 tokens correct)

The confusion matrix confirmed minimal cross-label confusion, with only 6 tokens misclassified across the entire validation set.

9. Error Analysis

9.1 Overall Error Summary

Metric	Value
Total Tokens	2876
Correctly Classified	2870
Misclassified	6
Overall Accuracy	99.79%

9.2 Per-Label Error Breakdown

Label	Total	Correct	Errors	Accuracy
ingredient	2107	2107	0	100.00%
quantity	411	409	2	99.51%
unit	358	354	4	98.88%

9.3 Misclassified Tokens

Token	Prev Token	True Label	Predicted	Context
to	10	unit	quantity	small 10 to 12 Green
into	cut	unit	quantity	French cut into 1 inch
into	cut	unit	quantity	cut into cm pieces
into	cut	unit	quantity	cut into pieces Salt
pinch	Dal	quantity	unit	Urad Dal pinch Asafoetida
cloves	Tomatoes	quantity	unit	Onion Tomatoes cloves Garlic

9.4 Error Analysis Insights

- Perfect Ingredient Classification: Zero misclassifications among 2107 ingredient tokens, demonstrating the model's strong ability to identify food items.
- Quantity Errors: "pinch" and "cloves" were misclassified as units. These words genuinely serve dual roles in cooking contexts (e.g., "a pinch of salt" vs "2 pinch").
- Unit Errors: "to" and "into" were misclassified as quantities. These contextual connectors appeared near numeric values (e.g., "10 to 12"), misleading the model.
- All errors involve inherently ambiguous tokens at entity boundaries where even human annotators might disagree.

10. Conclusion

This project successfully developed a CRF-based Named Entity Recognition model for extracting structured information from unstructured recipe text. Key findings:

- Data Preparation: 285 records were cleaned to 280 by removing 5 records with mismatched annotations. The data was split 70:30 into training (196) and validation (84) sets.
- Class Imbalance: The dominant ingredient class (~75%) was handled through inverse frequency weighting with additional penalisation (weight=0.67 vs quantity=8.65, unit=8.77).
- Feature Engineering: 25+ features per token were designed spanning core linguistic properties (spaCy POS/lemma/dependency), domain-specific detectors (regex patterns, keyword sets), and contextual features (neighbouring tokens, BOS/EOS markers).
- Model Performance: The CRF model achieved 99.94% training accuracy and 99.79% validation accuracy with only 6 misclassified tokens out of 2876.
- Error Analysis: All 6 errors involved genuinely ambiguous tokens ("to", "into", "pinch", "cloves") at entity boundaries. Zero ingredient misclassifications were observed.

The model is production-ready for recipe management systems, dietary tracking applications, and e-commerce platforms. Future improvements could include expanding the training data, adding bi-gram/tri-gram features, and implementing post-processing rules for known ambiguous tokens.